GRADE 9 INTEGRATED SCIENCE
CONTINUOUS ASSESSMENT TEST 2

TERM 1 | TIME: 1 HOUR 15 MINUTES | TOTAL: 50 MARKS

Name: _____ Adm. No.: _____

School: _____ Date: _____

INSTRUCTIONS

- Answer all questions.
- Show your working in calculations.
- Draw neat, labelled diagrams where necessary.
- The marks for each question are shown in brackets.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. State the number of protons, neutrons and electrons in a neutral atom of sodium-23. (2 marks)

_____

2. Define the term isotope. (2 marks)

_____

3. Write the electron arrangement of an atom with atomic number 13. (2 marks)

_____

4. Give TWO physical properties that make metals useful in making cooking utensils. (2 marks)

_____

5. Differentiate between an alloy and a pure metal. (2 marks)

_____

6. State TWO causes of temporary hardness of water. (2 marks)

_____

7. Name TWO methods that can be used to soften hard water. (2 marks)

_____

8. State TWO functions of the placenta during pregnancy. (2 marks)

_____

9. Give TWO differences between pollination and fertilisation in flowering plants. (2 marks)

_____

10. State TWO ways in which organisms in an ecosystem depend on one another. (2 marks)

_____

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Atomic structure and isotopes (8 marks)

(a) An atom has atomic number 17 and mass number 35. Determine the number of protons, electrons and neutrons in the atom. (3 marks)

_____

(b) Explain why atoms are electrically neutral. (2 marks)

_____

(c) Chlorine has two isotopes, chlorine-35 and chlorine-37. State what is the same and what is different in these isotopes. (3 marks)

_____

12. Metals, alloys and water (8 marks)

(a) Explain TWO reasons why alloys are often preferred to pure metals for making tools or machine parts. (4 marks)

_____

(b) Describe how temporary hardness of water can be removed by boiling. (2 marks)

_____

(c) State TWO disadvantages of hard water in the home. (2 marks)

_____

13. Reproduction in plants and animals (7 marks)

(a) Distinguish between self-pollination and cross-pollination. (2 marks)

_____

(b) Describe THREE events that occur after pollen grains land on a suitable stigma, leading to fertilisation. (3 marks)

_____

(c) State TWO functions of the amniotic fluid in a developing human baby. (2 marks)

_____

14. Interdependence of life (7 marks)

(a) Define a food chain. (2 marks)

_____

(b) Construct a food chain using the following organisms: grass, hawk, grasshopper and frog. (2 marks)

_____

(c) Explain THREE ways human activities can disrupt the balance of an ecosystem. (3 marks)

_____

MARKING SCHEME – GRADE 9 INTEGRATED SCIENCE CAT 2

Total: 50 marks

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Sodium-23 has 11 protons, 12 neutrons and 11 electrons. Award 1 mark for the correct proton/electron count and 1 for neutrons.

2. Isotopes are atoms of the same element with the same number of protons (same atomic number) but different numbers of neutrons/mass numbers. (2)

3. 2, 8, 3. (2)

4. Any TWO: good conductors of heat; strong; durable; malleable; corrosion-resistant where appropriate; easy to shape. (1 each)

5. A pure metal consists of one metallic element, whereas an alloy is a mixture of a metal with one or more other elements. (2)

6. Temporary hardness is commonly caused by dissolved calcium hydrogencarbonate and magnesium hydrogencarbonate. (1 each)

7. Any TWO: boiling (for temporary hardness); distillation; ion-exchange treatment; suitable water-softening methods. (1 each)

8. Any TWO: supplies oxygen and nutrients to the foetus; removes carbon dioxide and metabolic wastes; produces some hormones; transfers some maternal antibodies. (1 each)

9. Pollination is transfer of pollen from anther to stigma; fertilisation is fusion of male and female gamete nuclei. Pollination occurs before fertilisation. Award 1 mark for each clear difference.

10. Any TWO: feeding relationships; pollination; seed dispersal; shelter/habitat; decomposition and nutrient cycling; predator-prey relationships. (1 each)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Atomic structure and isotopes

(a) Protons = 17 (1); electrons = 17 for a neutral atom (1); neutrons = 35 − 17 = 18 (1).

(b) A neutral atom has equal numbers of positively charged protons and negatively charged electrons, so the charges balance. (2)

(c) Same: 17 protons/atomic number and chemical identity (1). Different: number of neutrons and mass number (2). Chlorine-35 has 18 neutrons; chlorine-37 has 20 neutrons.

12. Metals, alloys and water

(a) Any TWO explained, 2 marks each: alloys may be harder/stronger; may resist corrosion better; may be more wear-resistant; may have properties better suited to a particular use. (4)

(b) Boiling decomposes dissolved calcium/magnesium hydrogencarbonates to form insoluble substances that can settle and be removed by filtration/decanting. (2)

(c) Any TWO: forms scale in kettles/boilers; reduces soap lather and wastes soap; blocks pipes; causes deposits on utensils. (1 each)

13. Reproduction in plants and animals

(a) Self-pollination: pollen is transferred to the stigma of the same flower or another flower on the same plant. Cross-pollination: pollen is transferred to a flower on a different plant of the same species. (1 each)

(b) Any THREE in correct sequence: pollen grain germinates on stigma; pollen tube grows down the style; male nucleus/gamete travels through the tube; pollen tube enters the ovule; male and female nuclei fuse. (1 each)

(c) Any TWO: cushions/protects foetus from mechanical shock; allows movement; helps maintain a stable temperature; prevents drying and reduces friction. (1 each)

14. Interdependence of life

(a) A food chain is a sequence showing how energy/food is transferred from one organism to another through feeding. (2)

(b) Grass → grasshopper → frog → hawk. (2: correct sequence and arrows)

(c) Any THREE explained: deforestation destroys habitats; pollution poisons organisms; overfishing/hunting reduces populations; excessive pesticide use harms non-target organisms; burning/bush fires destroy habitats; introducing invasive species disrupts food webs. (1 each)

Teacher's note: This is original teacher-prepared assessment material. Confirm that the relevant Grade 9 sub-strands have been taught before administering. Award method marks for scientifically correct explanations and accept equivalent correct responses.